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1 Class theory: axioms and definitions

Constructor 1.1. Membership predicate.
If A and B are classes, then $(A \in B)$ is a formula.

Constructor 1.2. Subclass predicate.
If A and B are classes, then $(A \subseteq B)$ is a formula.

Definition 1.1. Subclass predicate.

$$(A \subseteq B) \ \longleftrightarrow \ (\forall x \in A : x \in B)$$

If the variant of logic being used doesn't have equality, then add the following constructor and definition.

Constructor 1.3. Equality.
If A and B are classes, then $(A = B)$ is a formula.

Definition 1.2. Equality. (in case if the 1st order logic is given without equality) If Y is not a free variable neither in A nor in B, then:

$$(A = B) \ \longleftrightarrow \ (\forall Y : A \in Y \ \longleftrightarrow \ B \in Y)$$

1.1 Axiom of extensionality

Axiom 1. Extensionality.

$$(A = B) \ \longleftrightarrow \ ((A \subseteq B) \wedge (B \subseteq A))$$

Axiom. Extensionality. This is an other variant of the definition, with requirement that X is a variable different from both A and B). If we understand A and B as variables, then it is enough; if as terms, then x should not be free neither in A nor in B.

$$(A = B) \ \longleftrightarrow \ (\forall X : X \in A \ \longleftrightarrow \ X \in B)$$

Constructor 1.4. Set predicate.
If A is a class, then (A set) is a formula.

Definition 1.3. Set predicate. If Y is not a free variable in X, then:

$$(X \text{ set}) \ \longleftrightarrow \ (\exists Y : X \in Y)$$

Constructor 1.5. If φ is a formula, then $\{x|\varphi\}$ is a class.
(It is Mosrse-Kelley class theory, since there are no restrictions on φ, which is the only difference between MK and NBG.)

1.2 Axiom of class comprehension

Axiom 2. (OLD Class comprehension)

Axiom 3. (NEW Class comprehension)

$$(X \in \{X|\varphi\}) \longleftrightarrow ((X \text{ set}) \wedge \varphi)$$

$$(X \in \{X|\varphi\}) \longleftrightarrow ((X \text{ set}) \wedge \varphi)$$

Let's substitute X with some variable Y, which is different from X.

$$(Y \in \{X|\varphi\}) \longleftrightarrow ((Y \text{ set}) \wedge \varphi[Y/X])$$

Do the same step but for $\varphi[X/W]$.

$$(X \in \{X|\varphi[X/W]\}) \longleftrightarrow ((X \text{ set}) \wedge \varphi[X/W])$$

$$(Y \in \{X|\varphi[X/W]\}) \longleftrightarrow ((Y \text{ set}) \wedge \varphi[X/W][Y/X])$$

Is the following statement true:
"If $\varphi[X/W]$ is defined(sic!), and if $\varphi[X/W][Y/X]$ is defined(btw the first follows from th second), and if (Y is not a free variable of φ) then $\varphi[X/W][Y/X]$ is equal to $\varphi[Y/W]$"?
...

Now, let's prove "old class comprehesion" statement.

Theorem 1.1. ()
If Y is not a free variable of φ, then:

$$(Y \in \{X|\varphi[X/W]\}) \longleftrightarrow ((Y \text{ set}) \wedge (\varphi[Y/W]))$$

Proof.
($\longrightarrow$)
Assume $Y \in \{X|\varphi[X/W]\}$.

...
Done.

1.3 Empty class

Constructor 1.6. Empty class.
The empty set symbol $\emptyset$ is a constant closed term.

Definition 1.4. Empty class.

$$\emptyset = \{x|\bot\}$$

Definition 1.5. Empty set predicate.

$$(A \text{ empty}) \longleftrightarrow (A = \emptyset)$$

Definition 1.6.

$$(A \text{ nonempty}) \longleftrightarrow \neg(A \text{ empty})$$

1.4 Axiom of empty set

Axiom 4.

$$(\emptyset \text{ set})$$

1.5 Universal class

Definition 1.7. Universal class (constant V).

$$V := \{x|\top\}$$

1.6 Union

Definition 1.8. Union of an arbitrary class A.

$$\bigcup A = \{x|\exists a \in A : x \in a\}$$

1.7 Intersection

Definition 1.9. Intersection of an arbitrary class A.

$$\bigcap A = \{x|\forall a \in A : x \in a\}$$

1.8 Powerclass

Definition 1.10. Powerclass. (by class comprehension)

$$\mathcal{P}X = \{x|x \subseteq X\}$$

1.9 Axiom of heredity

Axiom 5. Heredity.

$$(A \text{ set}) \wedge (B \subseteq A) \to (B \text{ set})$$

1.10 Axiom of powerset

Axiom 6. Powerset. (weak version, but it is okey, since together with axiom of heredity it is equivalent to the strong version.)

$$(A \text{ set}) \to (\mathcal{P}A \text{ set})$$

1.11 Axiom of regularity / Epsilon-induction

Axiom 7. $\in$-induction.

$$(\mathcal{P}\Psi \subseteq \Psi) \implies (V \subseteq \Psi)$$

1.12 Axiom of limitation of size

Axiom 8. Limitation of size (of a set).

$$\neg(X \text{ set}) \longleftrightarrow (\exists T : (T \text{ function}) \wedge (V \subseteq T[X]))$$